

NenuFAR – ES17

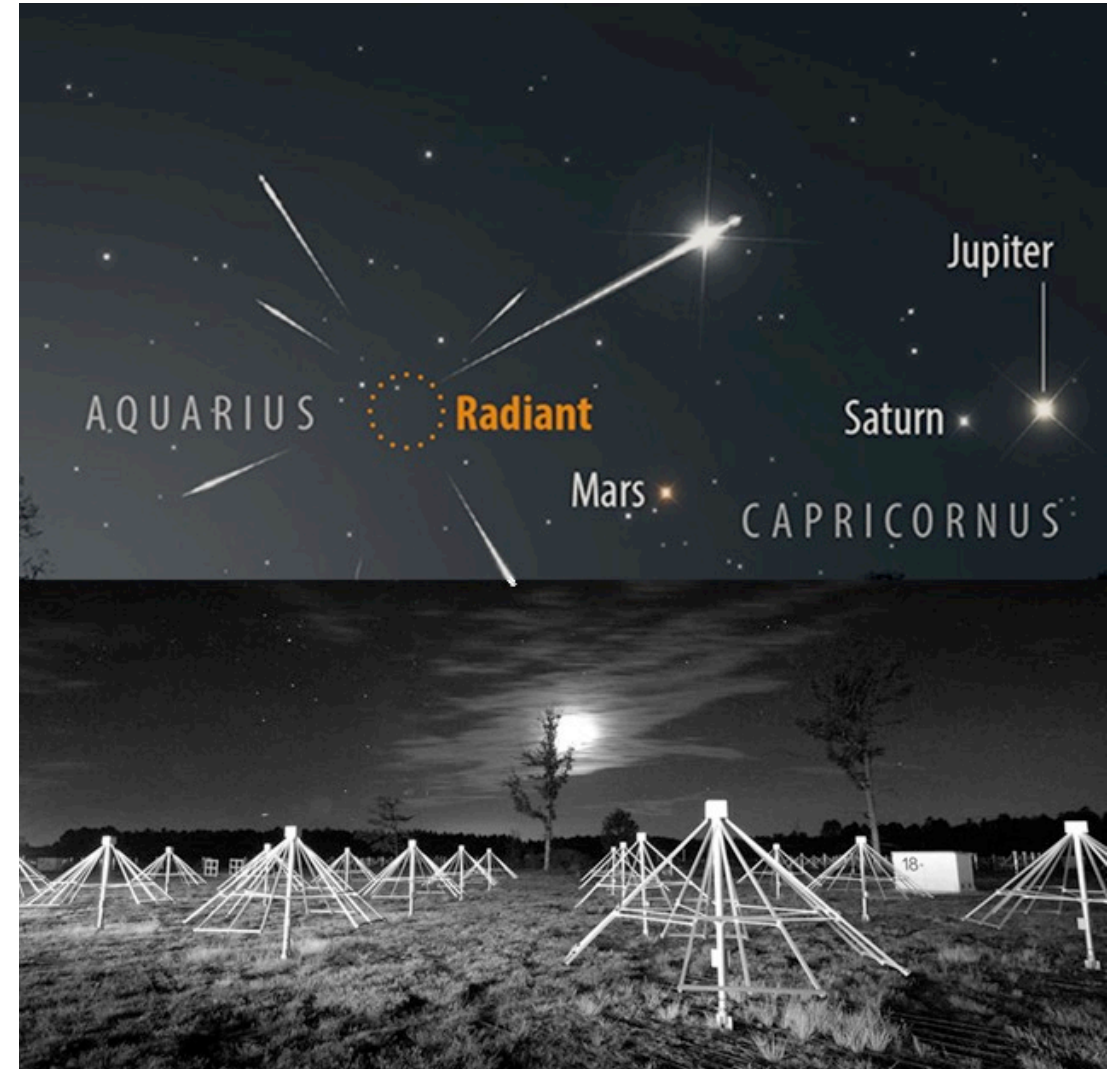
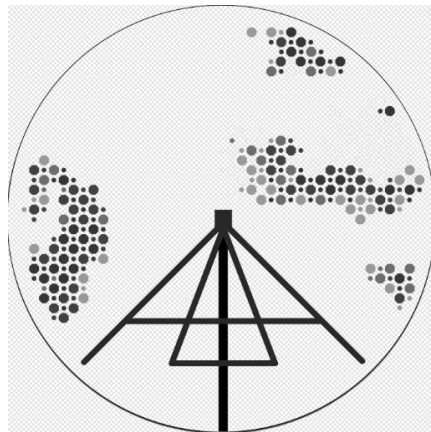
Amateur radioastronomy

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NENUFAR – ES17 – Amateur Radioastronomy

1. Deployment stage: activities
2. Learning stage: "amateur" activities
3. Main activities targeting meteoritic showers

Workshop 2021



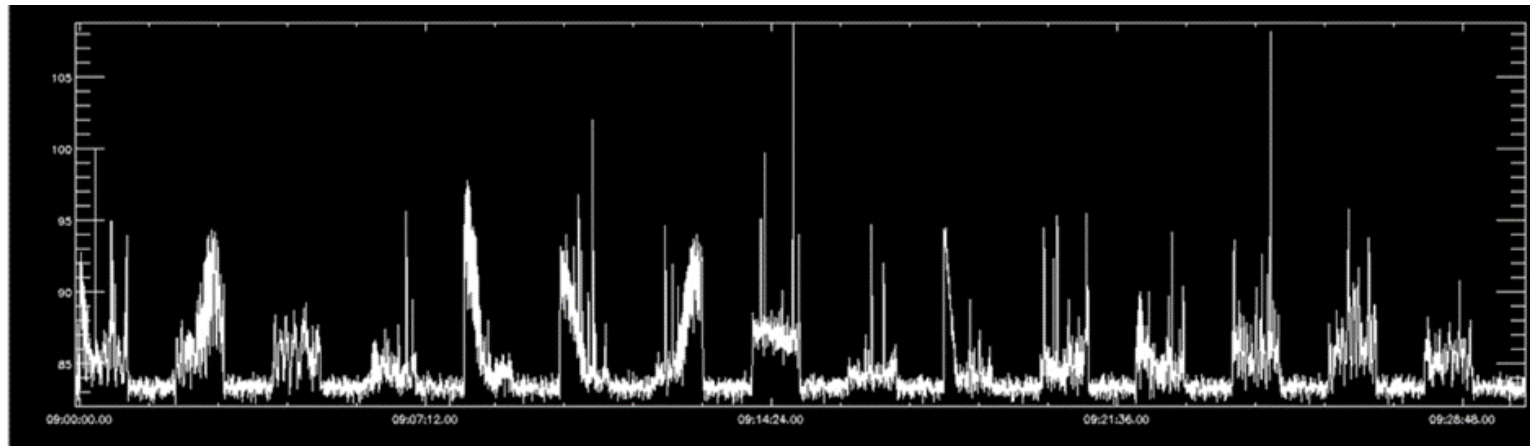
ES17 - NenuFAR deployment stage

Supervised by Laurent Denis and Julien Girard

- ❑ Help in evaluating elevation-based angular pointing
 - Use of terrestrial beacons ("HAM" F5ZKY)
 - Dedicated transmission (F2CT HAM station) from the Pyreneans
 - CW¹ ON/OFF signals (specific dedicated timing)
 - Dedicated frequencies (lines)

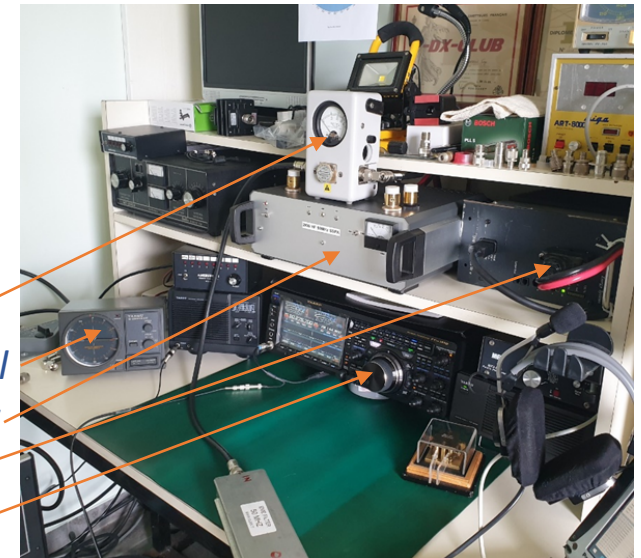


F2CT HAM station



- ✓ Received signals at NenuFAR: 60s ON / 60s OFF (30mn sequence)
- ✓ Results used by Nançay (performance estimation at low elevation)

Power meter
Beaming control
Power Amplifier
Power Supply
Tx Synthesizer

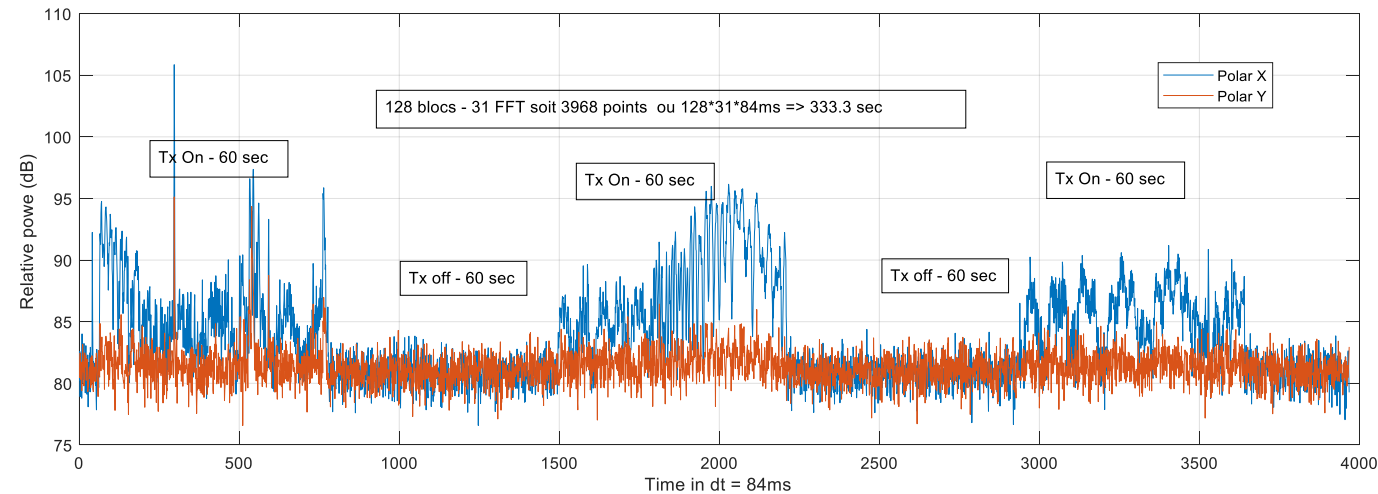


¹ CW : Continuous Wave

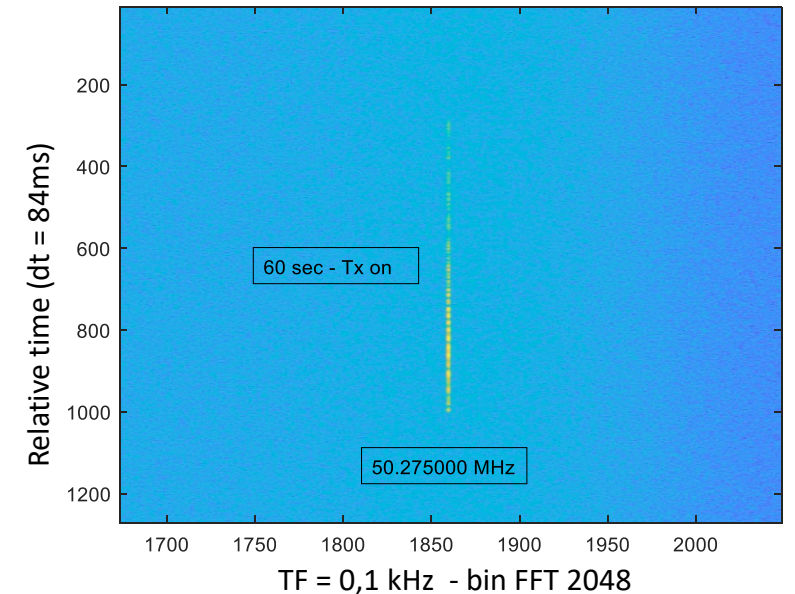
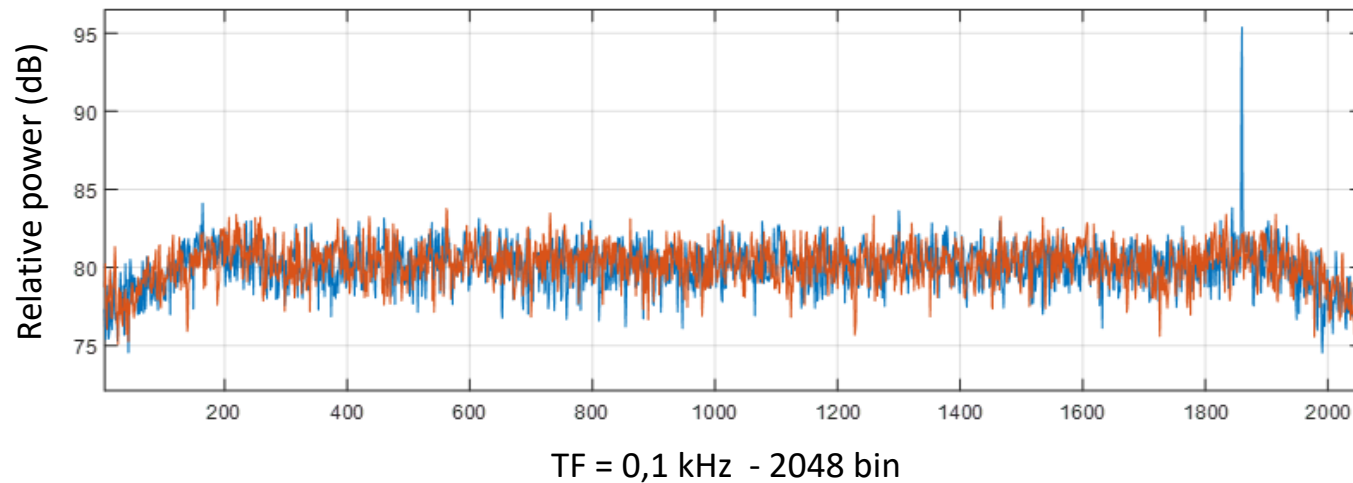
ES17 - NenuFAR deployment stage

Supervised by Laurent Denis and Julien Girard

- ❑ Estimation of the NenuFAR frequency plan
 - Same procedure...
 - Highlighting a frequency shift of a half-channel ($195\text{kHz}/2$)
 - Allowed the frequency plan correction (implemented by Nançay)

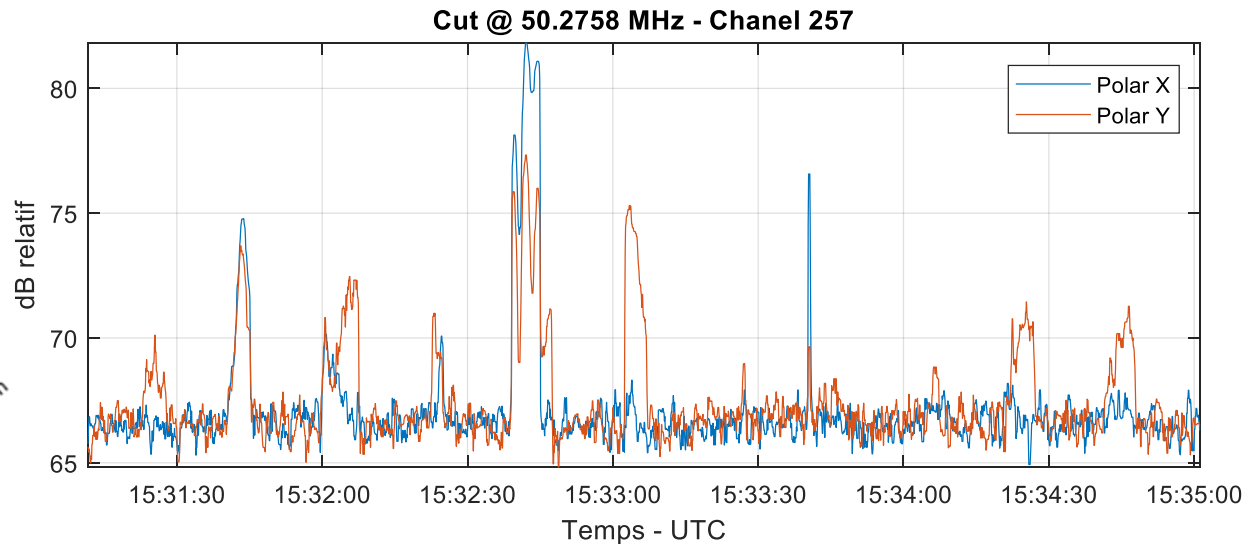
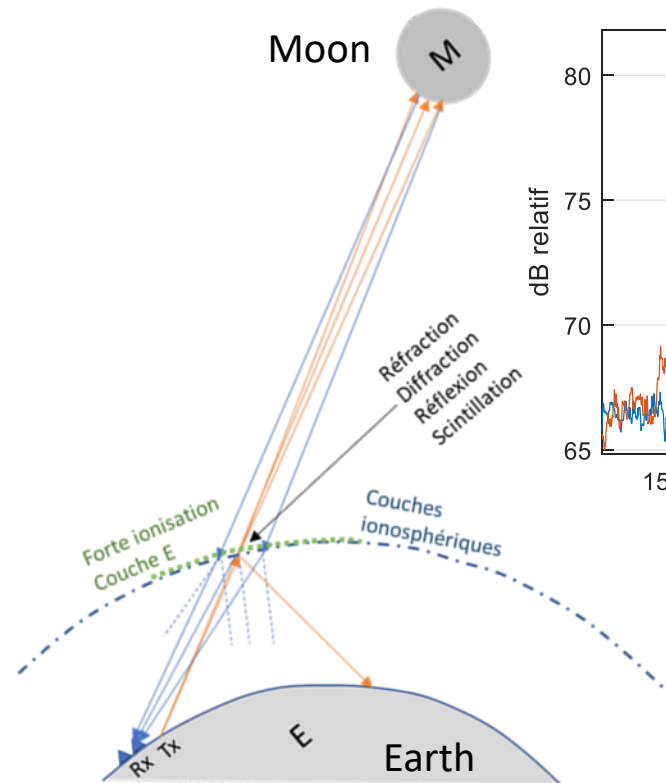


Channel 257 - TF = 0,1 kHz - 2048 bin



ES17 - NenuFAR deployment stage

- ❑ Evaluation of the SNR budget (Moon (EME) / ISS reflection)
- ❑ Observation of ionospheric disturbances
 - Transmission (from "HAM" station)
 - 10s ON / 20s OFF CW signals (high power antenna, 800 W peak @ antenna input)

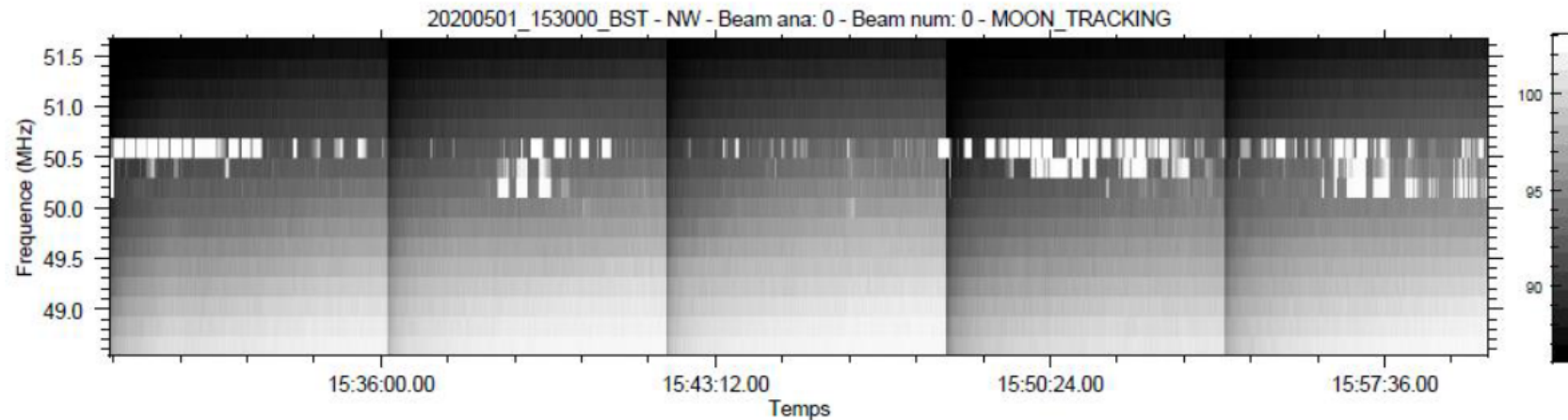


50 MHz antenna pointed at the moon

- Outbound signal received at NenuFAR:
 - Expected SNR: + 6 to 7 dB => 74 to 75 dB on the plot
 - Strong ionospheric perturbations (outbound and inbound) modulating the signal amplitude

ES17 - NenuFAR deployment stage

- ❑ Contributions to RFI and atmospheric propagation monitoring
 - Amateur Radio digital communication channels



- Radio-amateurs use reflections on the ionized layers of the ionosphere (E layer) to establish long distance links
- One man's loss (radio astronomical signals disturbances) makes another man's gain (long distance links)

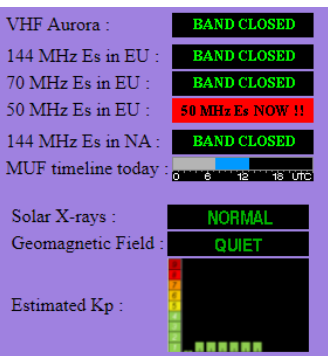
- The propagation of amateur bands is monitored continuously

6m band (50 MHz) openings:

<http://www.gooddx.net/>

Real time radio links map on 10, 6, 4 and 2m bands

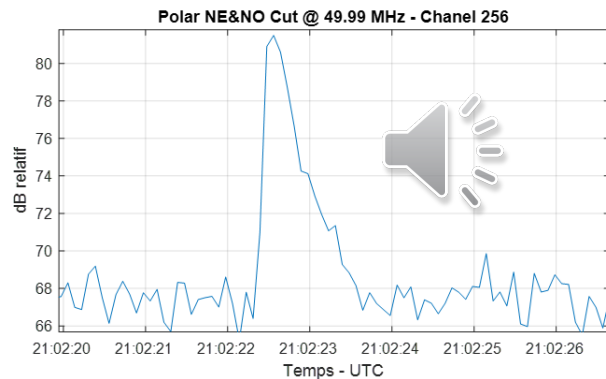
Live MUF V7 Mapping



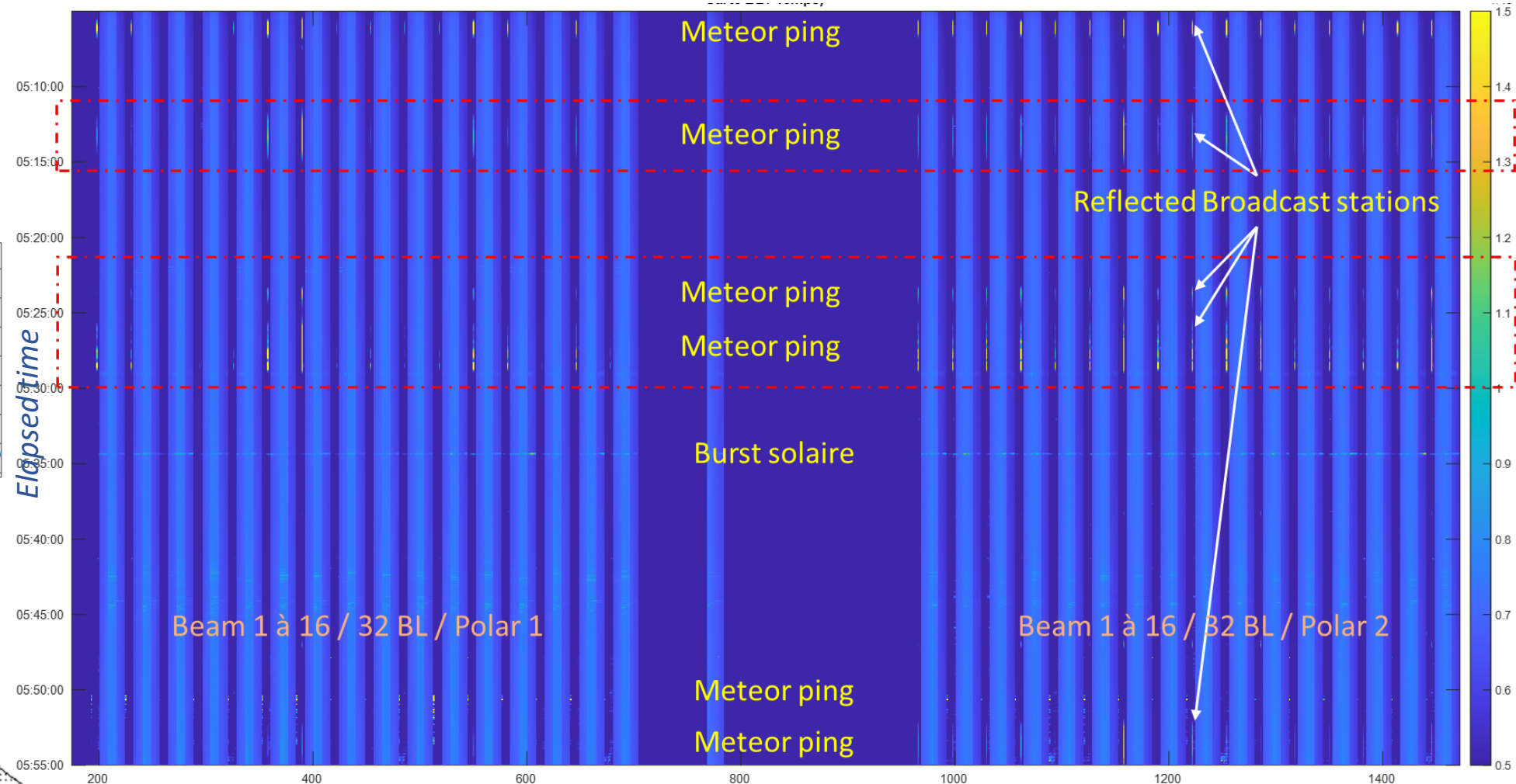
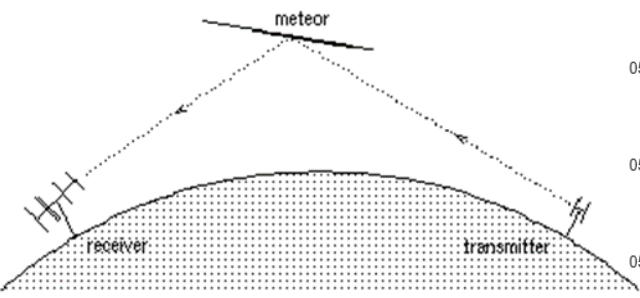
ES17 - NenuFAR deployment phase

- ❑ Contributions to RFI and propagation monitoring
 - RFI from terrestrial broadcasting stations reflected on meteor trails

Beamlets NenuFAR
Survey of the "meteors pings"
(Radio signals reflected on the meteor trails)



Typical underdence meteor



ES17 - Learning

Supervised by Julien Girard

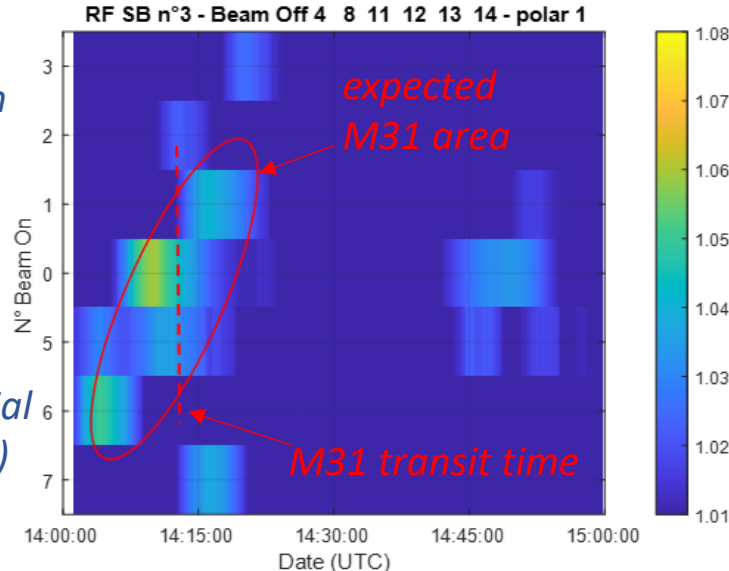
☐ Observations of M31 and "powerful" radio sources (A-Team)

➤ Development of analysis algorithms for NenuFAR data (BST and TF)

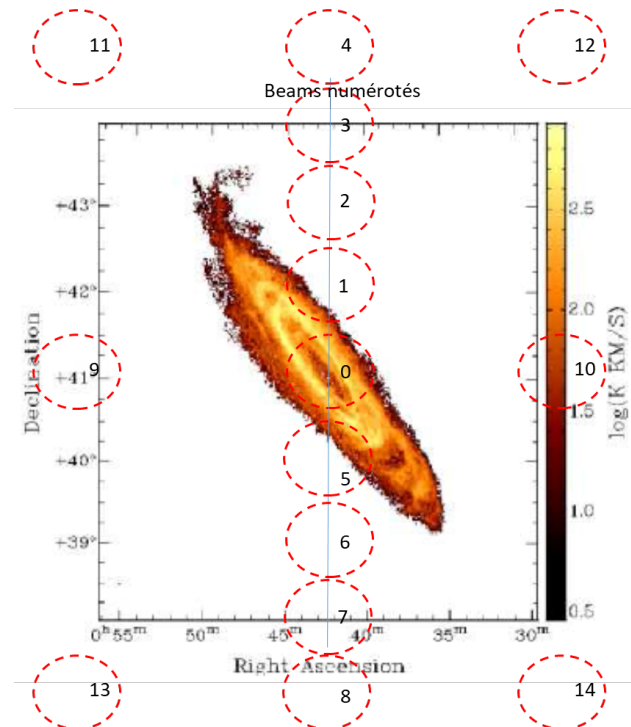
✓ Findings:

- Insufficient knowledge of the scientific models applied to these observations
- Insufficient knowledge of the processing pipelines used at Nançay
- To be further studied (workshop attendance)

*Observation of M31 in transit multi-beam mode.
Strong ionospheric disturbances.
Processing and realignment (equatorial projection in progress)*



✓ "A-Team" observation ongoing



Setup

- ✓ 9 beams dedicated to the M31 transit (Beams ON 0 to 8)
- ✓ 6 extra beams (Beams OFF)
- ✓ 6 RF sub-bands @ 33, 43, 53, 63, 73 & 83 MHz
- ✓ 8 channels / sub-band => 48 freq. channels
- ✓ 15 beams * 48 ch. => 720 beamlets (max 768)
- ✓ Dynamic spectrum => $t_f: df=12.20$
 $dt=84.0$ (Hamming FFT window)

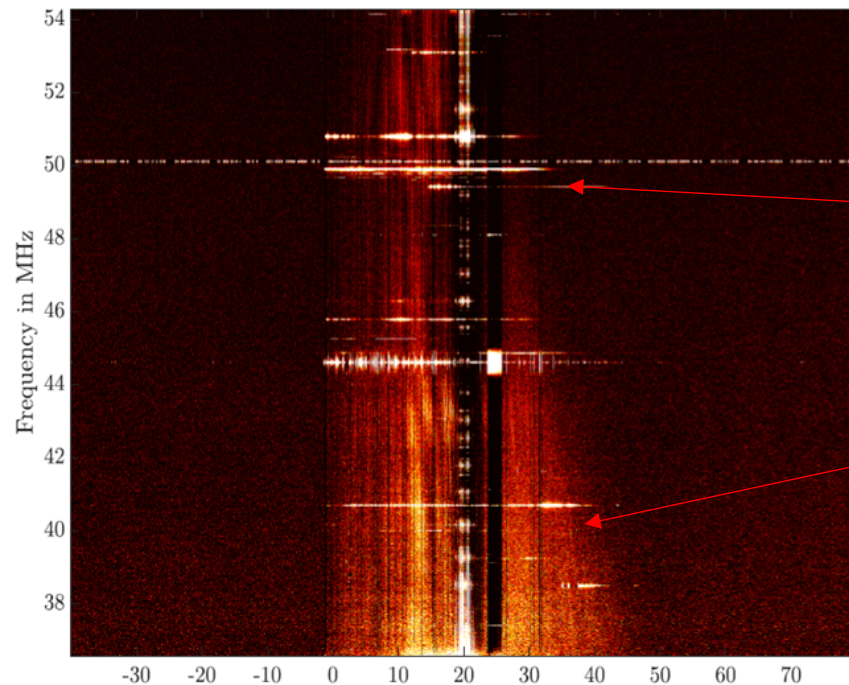
ES17 - Meteorite showers

Supervised by Julien Girard

Goal:

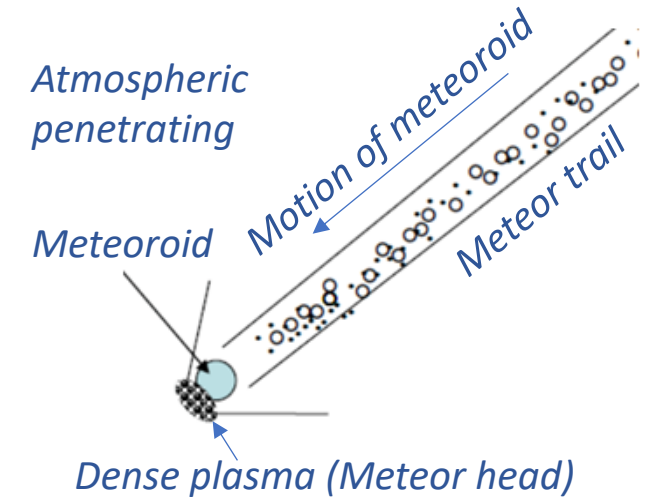
- ❑ Highlight the plasma radiation of meteors during their penetration in the ionosphere (see bellow)
- ❑ First observations made by Obenberger using the LWA1 array

https://science.nrao.edu/science/meetings/2016/15th-synthesis-imaging-workshop/Obenberger_SISS16_Radio_Meteors.pdf



Passive radio radiation (lines) on meteorite trail

Broadband plasma radiation

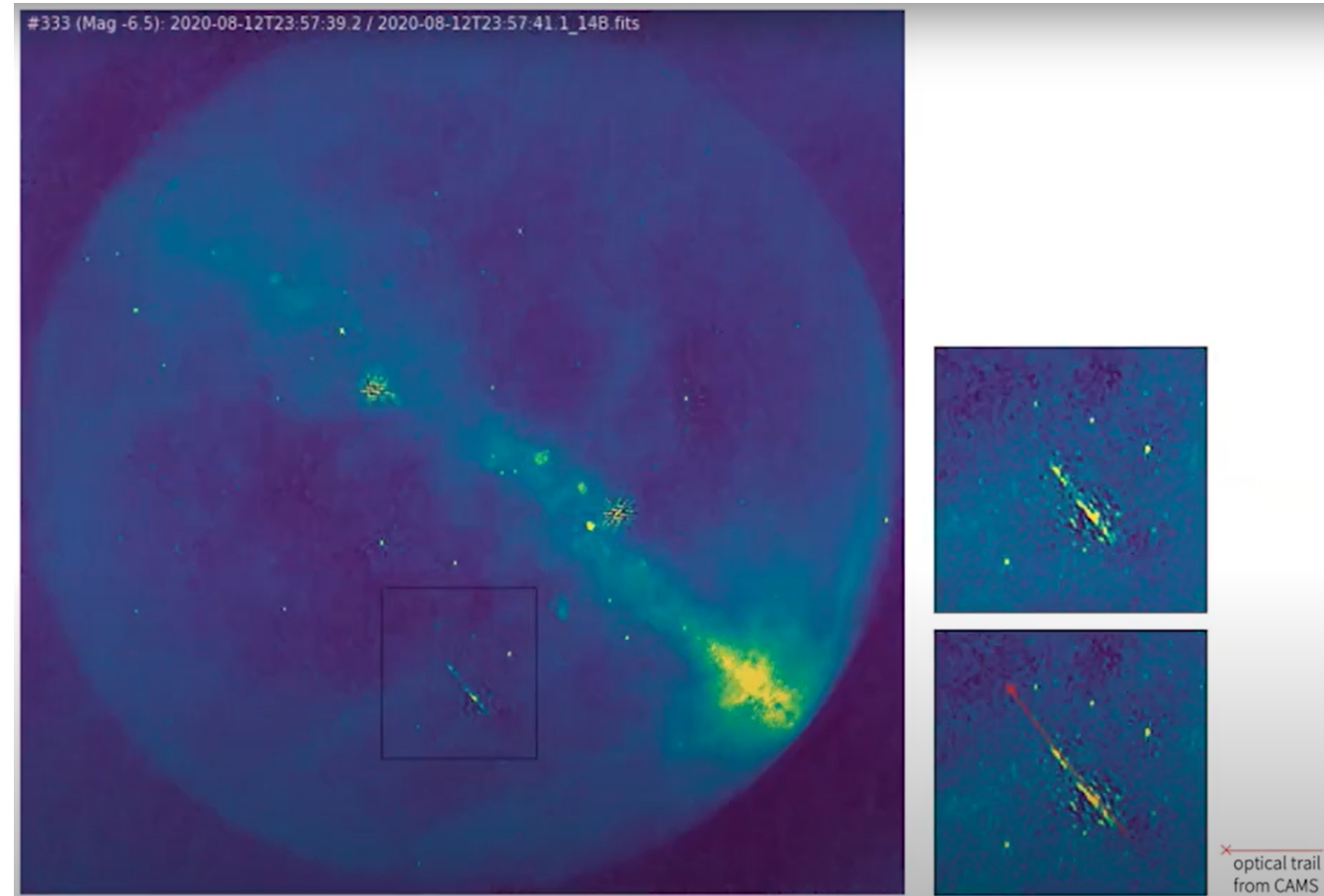


Meteor trail (optical)

ES17 - Meteorite showers

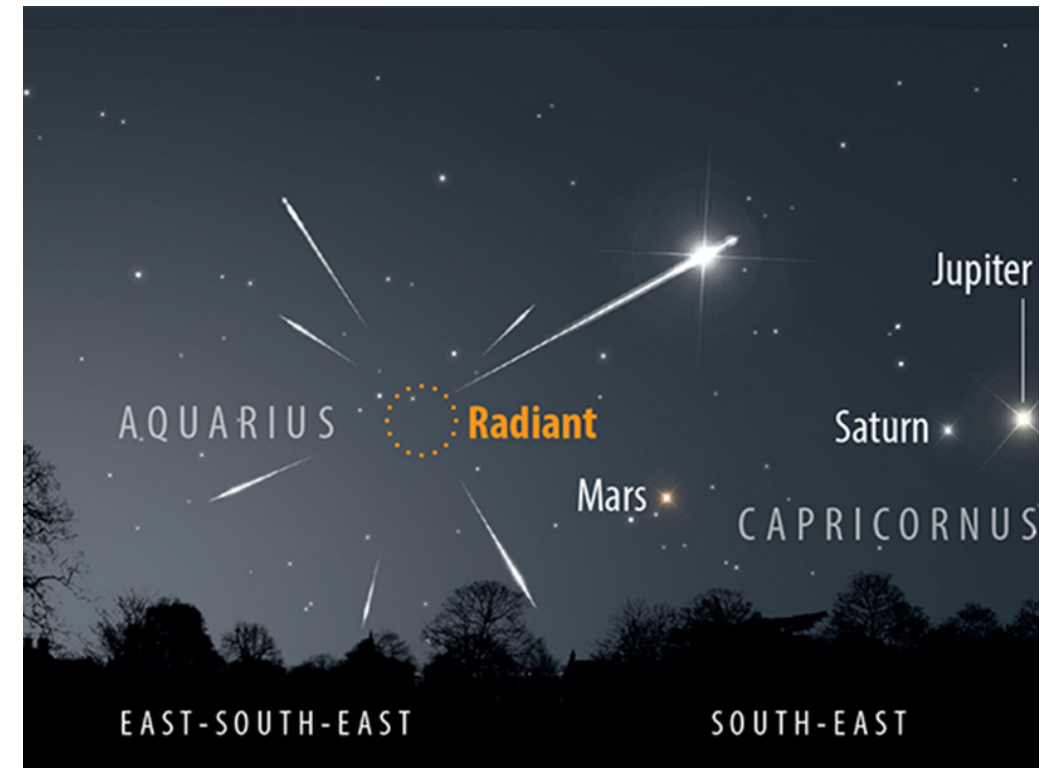
- ❑ Last update : LOFAR team presentation at IMC2021 conference (October)
- ❑ LOFAR has detected broadband radiation during meteor streak observations
https://www.youtube.com/watch?v=8JgBUrEEzzo&list=PLEypPv_8eK00plVBKu21xFrymUaRiZQSc&ab_channel=InternationalMeteorOrganization
- ❑ In a first assessment, the LOFAR team attributes this signal to a possible reflection of the galactic center radiation on the meteor trails

To be continued...



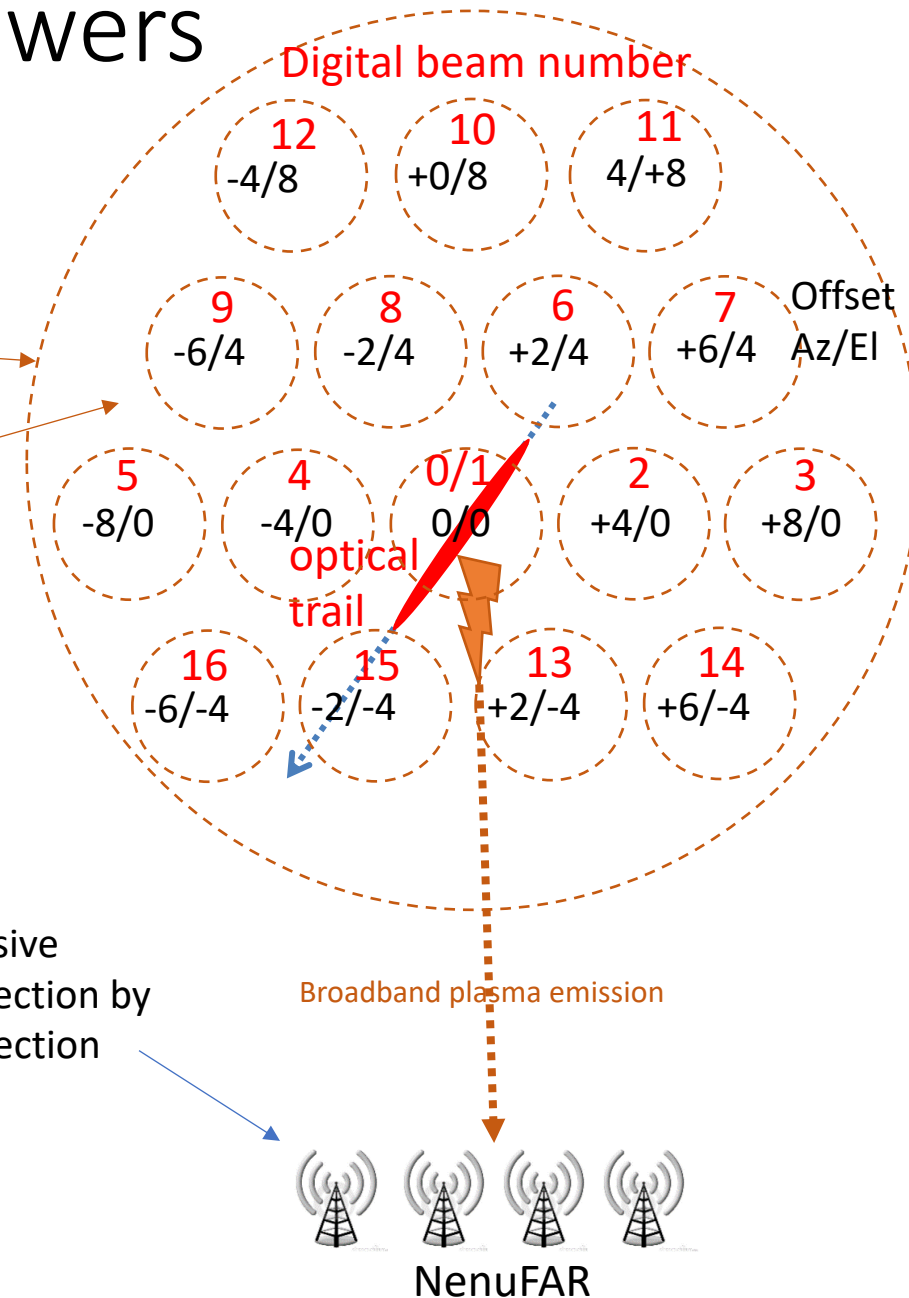
ES17 - Meteorite showers

- ❑ The 2020 observations made by ES17 have made it possible to redefine an observation configuration for late 2021 (in progress) and early 2022
- ❑ Main limitations:
 - ✓ We know the overall coordinates of arrival of the shower (radiant), but we don't know precisely where an event will happen
 - ✓ Same for the time, we know the calendar period, but not the exact time of atmospheric penetration
 - ✓ Each event duration is very short, while the time window during which the event will occur is fairly large. In contrast, NenuFAR slots are shorts and limited
 - ✓ ES17 is a small team



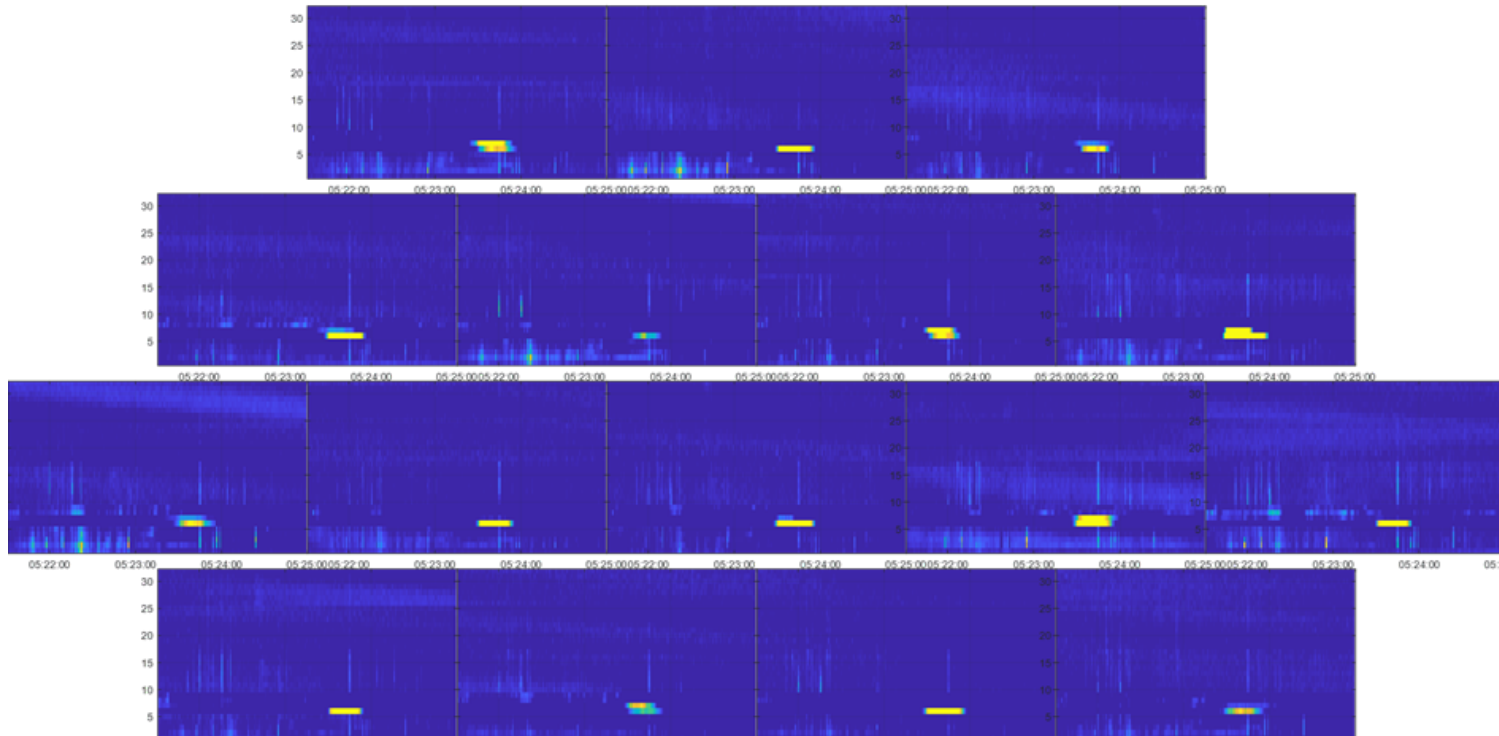
ES17 - Meteorite showers

- ❑ The new configuration takes into account the main restrictions.
- ❑ Configuration:
 - ✓ 1 analog beam (56 mini-arrays pointing to the shower radiant) => **maximum sensitivity**
 - ✓ 16 TF digital beams (1 to 16) to obtain a large angular coverage => **highlight a likely direction**
 - ✓ 4 sub-bands @ 26, 38, 48, 52 MHz / 8 channels => **highlight the broadband emission**
 - ✓ $Dt = 84\text{ms}$, suited to an event duration
 - ✓ 1 digital beam (0) ($tf = 0.1\text{kHz}$) centered to detect the ground beacons signals reflected by the meteor trails => **detect the moments of interest**
 - ✓ 5 hours of observations centered around the expected peak of activity, with 5 fixed pointing (1 re-pointing per hour) to avoid the "glitch" of beam switching, which exhibits the same transient nature as the sought signal



ES17 - Meteorite showers

- ❑ Observations for the main showers are in progress and requested for the next semester according to the IMO* schedule
- ❑ Development of processing tools in progress (use of On/Off beam) according to the 16 recorded digital beams and according to a time (instant of detection) / frequency (32 BL) / level after filtering / beam



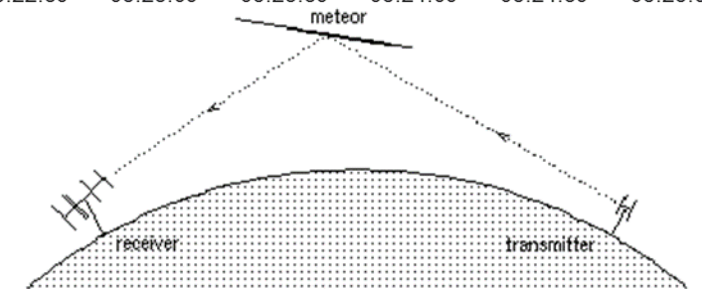
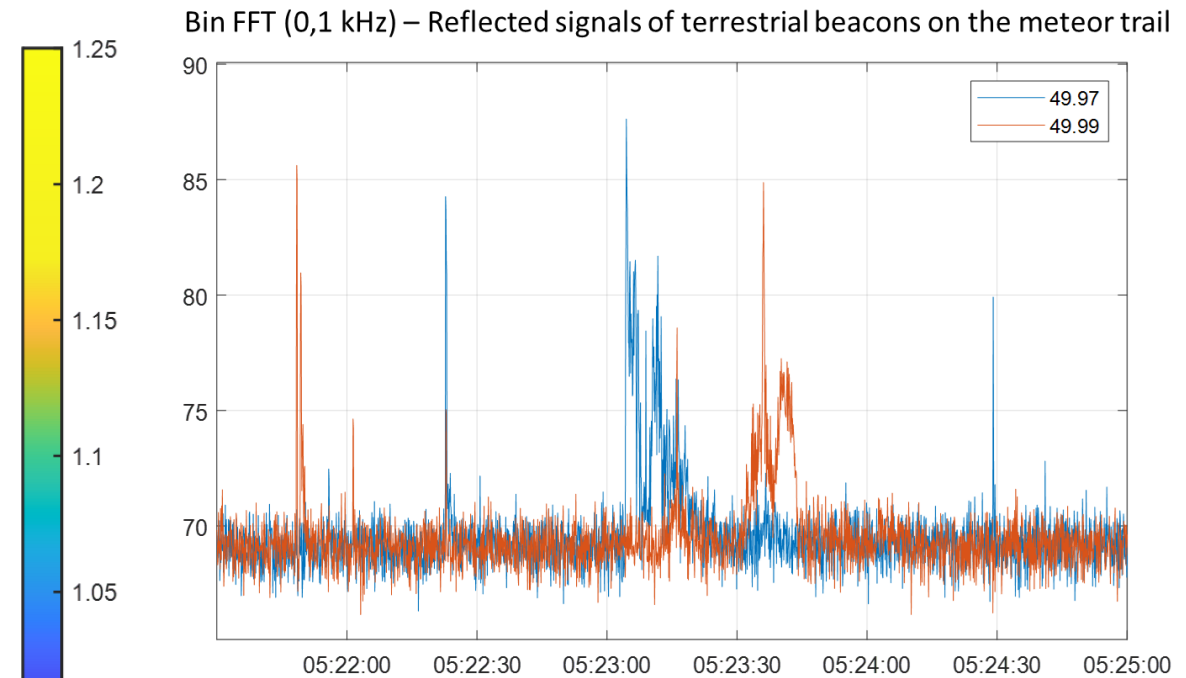
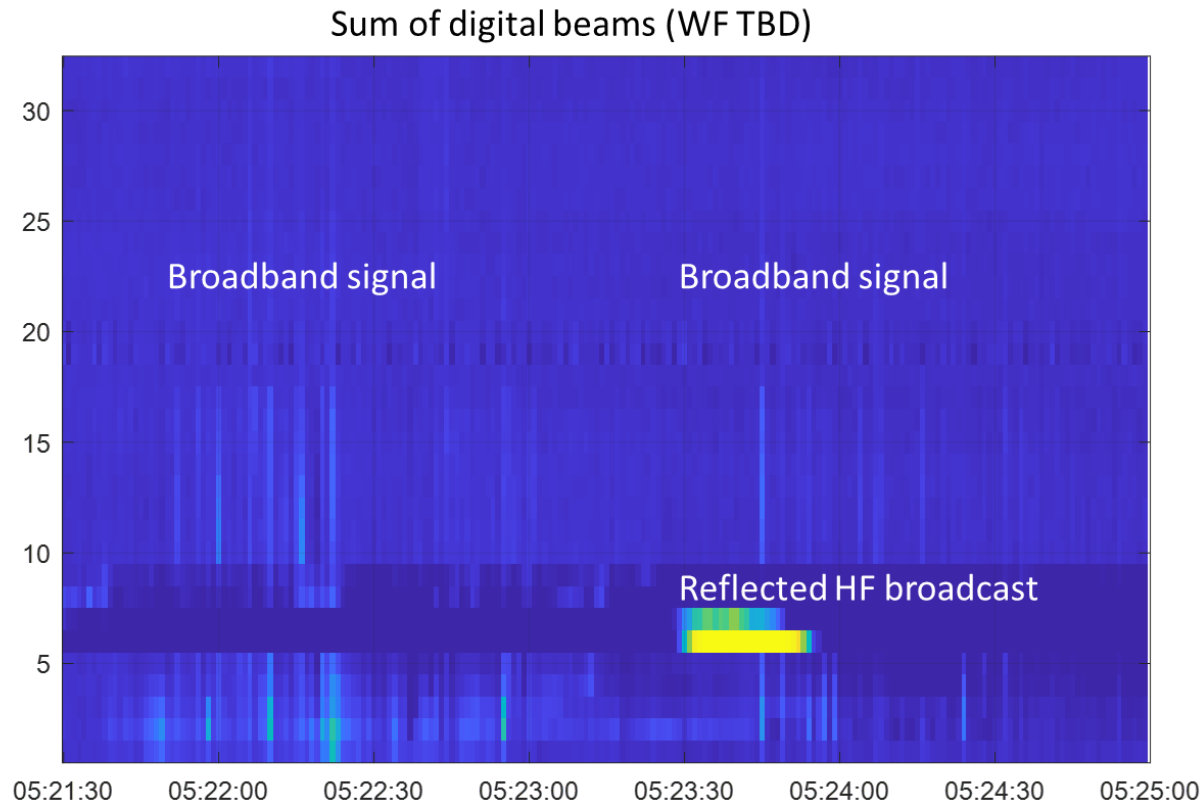
On this view, the signals have no favored direction of arrival.

The objective of the next observations and analyses is to visualize favored directions and their associated broadband emission detection.

*IMO : International Meteor Organisation (host of IMConference)

ES17 - Meteorite showers

- ❑ The moments of observation (analysis) are centered on the moments of passive detection of HF broadcast or ground beacons (on the dedicated channel)



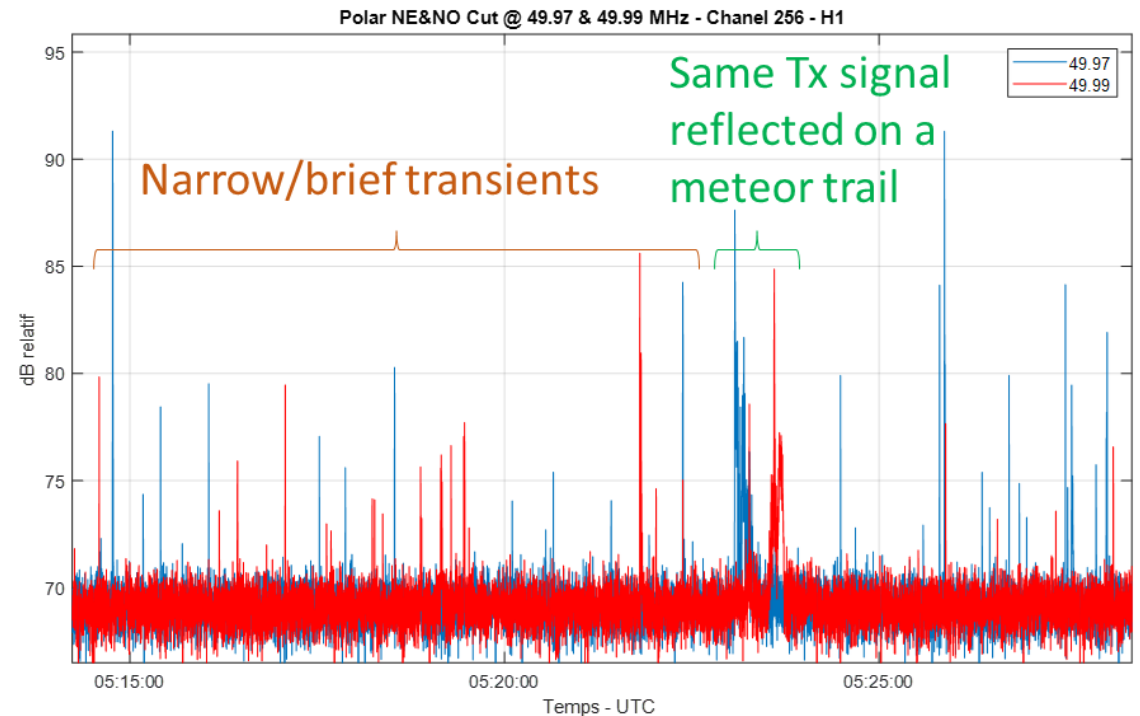
Next observations planned with TF waveform (84ms)

ES17 - Meteorite showers

To sum up:

- ❑ Observations of the main showers are in progress and requested for the next semester
- ❑ The TBD waveform (1s) was too long and has been replaced by a shorter TF waveform (84 ms)
- ❑ **Other signals** of shorter nature have been observed and their origin is so far unexplained (for ES17). These narrow band signals (line) are transmitted by the ground beacons (Belgian) and are probably from a reflection on an "atmospheric object". The peaks are frequent and present a significant SNR in $B=0.1\text{kHz}$ as shown in the graph opposite.

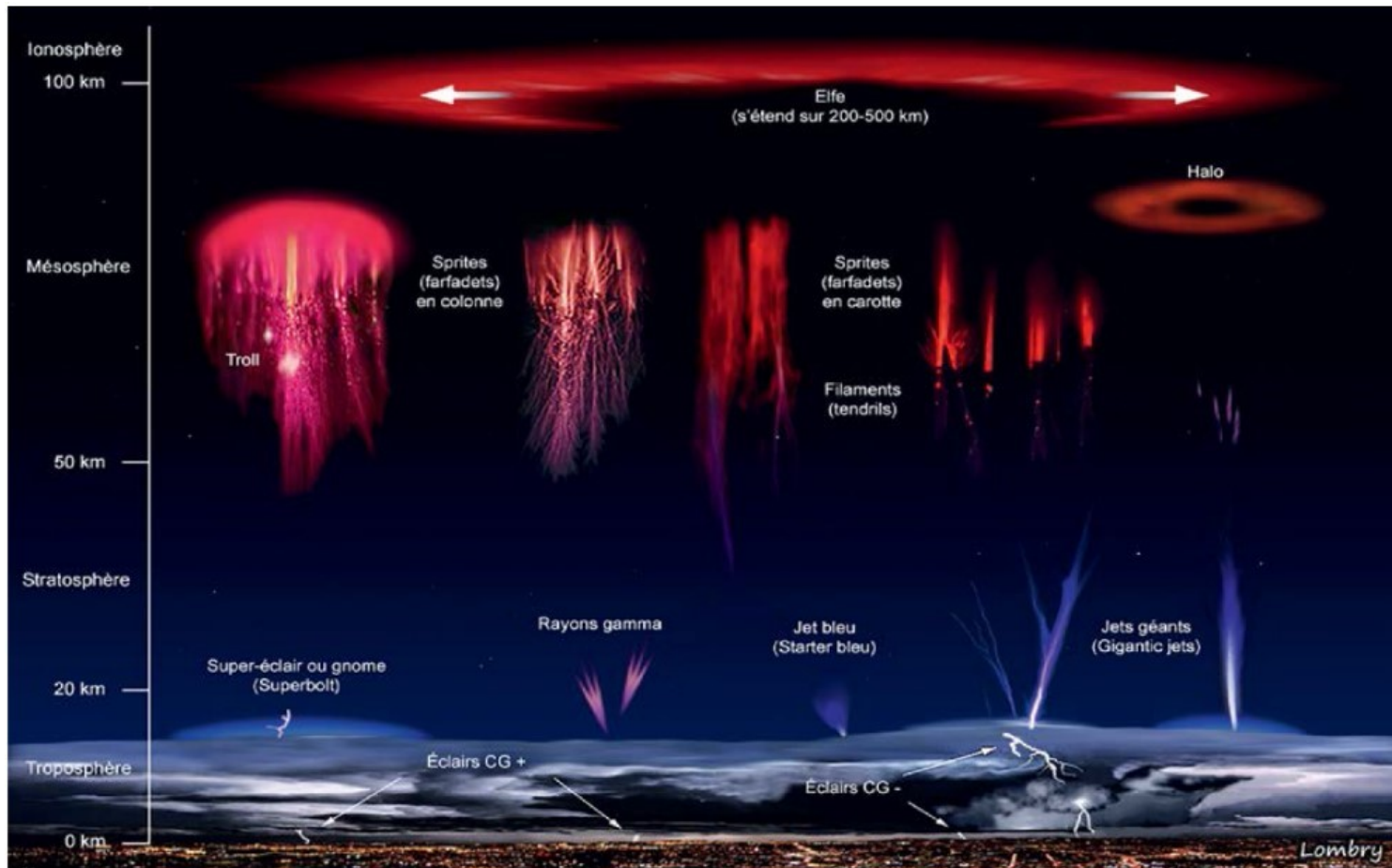
- ❑ Could this be related to electrostatic discharges ionizing the lower atmosphere? These signals are always present, very numerous and do not seem to be disturbed by some abnormal propagations linked to the Ionospheric layers (which disturb the meteorite observations)
- Could it be possible to record signals with a bandwidth lower than 0.1kHz, or an IQ recording of a single 195 kHz channel?



ES17 - Meteorite showers

Some examples of possible ionizing events (perspectives)

- ❑ TLE (Transient Luminous Events) such as red sprites, elves, blue jets, halo or trolls are upper atmospheric optical phenomena associated with thunderstorm



https://www.youtube.com/watch?v=vJ2jXI-MMI4&ab_channel=ObservatoireMidi-Pyr%C3%A9n%C3%A9es

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